

8 An Introduction to Metabolism 143

CONCEPT 8.1 An organism's metabolism transforms matter and energy 144

- Metabolic Pathways 144
- Forms of Energy 144
- The Laws of Energy Transformation 145

CONCEPT 8.2 The free-energy change of a reaction tells us whether or not the reaction occurs spontaneously 147

- Free-Energy Change, ΔG 147
- Free Energy, Stability, and Equilibrium 147
- Free Energy and Metabolism 148

CONCEPT 8.3 ATP powers cellular work by coupling exergonic reactions to endergonic reactions 150

- The Structure and Hydrolysis of ATP 150
- How ATP Provides Energy That Performs Work 151
- The Regeneration of ATP 153

CONCEPT 8.4 Enzymes speed up metabolic reactions by lowering energy barriers 153

- The Activation Energy Barrier 153
- How Enzymes Speed Up Reactions 154
- Substrate Specificity of Enzymes 155
- Catalysis in the Enzyme's Active Site 156
- Effects of Local Conditions on Enzyme Activity 157
- The Evolution of Enzymes 159

CONCEPT 8.5 Regulation of enzyme activity helps control metabolism 159

- Allosteric Regulation of Enzymes 160
- Localization of Enzymes Within the Cell 161



9 Cellular Respiration and Fermentation 164

CONCEPT 9.1 Catabolic pathways yield energy by oxidizing organic fuels 165

- Catabolic Pathways and Production of ATP 165
- Redox Reactions: Oxidation and Reduction 165
- The Stages of Cellular Respiration: *A Preview* 168

CONCEPT 9.2 Glycolysis harvests chemical energy by oxidizing glucose to pyruvate 170

CONCEPT 9.3 After pyruvate is oxidized, the citric acid cycle completes the energy-yielding oxidation of organic molecules 171

- Oxidation of Pyruvate to Acetyl CoA 171
- The Citric Acid Cycle 172

CONCEPT 9.4 During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis 174

- The Pathway of Electron Transport 174
- Chemiosmosis: The Energy-Coupling Mechanism 175
- An Accounting of ATP Production by Cellular Respiration 177

CONCEPT 9.5 Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen 179

- Types of Fermentation 180
- Comparing Fermentation with Anaerobic and Aerobic Respiration 181
- The Evolutionary Significance of Glycolysis 182

CONCEPT 9.6 Glycolysis and the citric acid cycle connect to many other metabolic pathways 182

- The Versatility of Catabolism 182
- Biosynthesis (Anabolic Pathways) 183
- Regulation of Cellular Respiration via Feedback Mechanisms 183



10 Photosynthesis 187

CONCEPT 10.1 Photosynthesis feeds the biosphere 188

CONCEPT 10.2 Photosynthesis converts light energy to the chemical energy of food 189

- Chloroplasts: The Sites of Photosynthesis in Plants 189
- Tracking Atoms Through Photosynthesis 189
- The Two Stages of Photosynthesis: *A Preview* 191

CONCEPT 10.3 The light reactions convert solar energy to the chemical energy of ATP and NADPH 192

- The Nature of Sunlight 192
- Photosynthetic Pigments: The Light Receptors 192
- Excitation of Chlorophyll by Light 195
- A Photosystem: A Reaction-Center Complex Associated with Light-Harvesting Complexes 195
- Linear Electron Flow 197
- Cyclic Electron Flow 198
- A Comparison of Chemiosmosis in Chloroplasts and Mitochondria 199

CONCEPT 10.4 The Calvin cycle uses the chemical energy of ATP and NADPH to reduce CO₂ to sugar 201

CONCEPT 10.5 Alternative mechanisms of carbon fixation have evolved in hot, arid climates 203

- Photorespiration: An Evolutionary Relic? 203
- C₄ Plants 203
- CAM Plants 205

CONCEPT 10.6 Photosynthesis is essential for life on Earth: a review 206